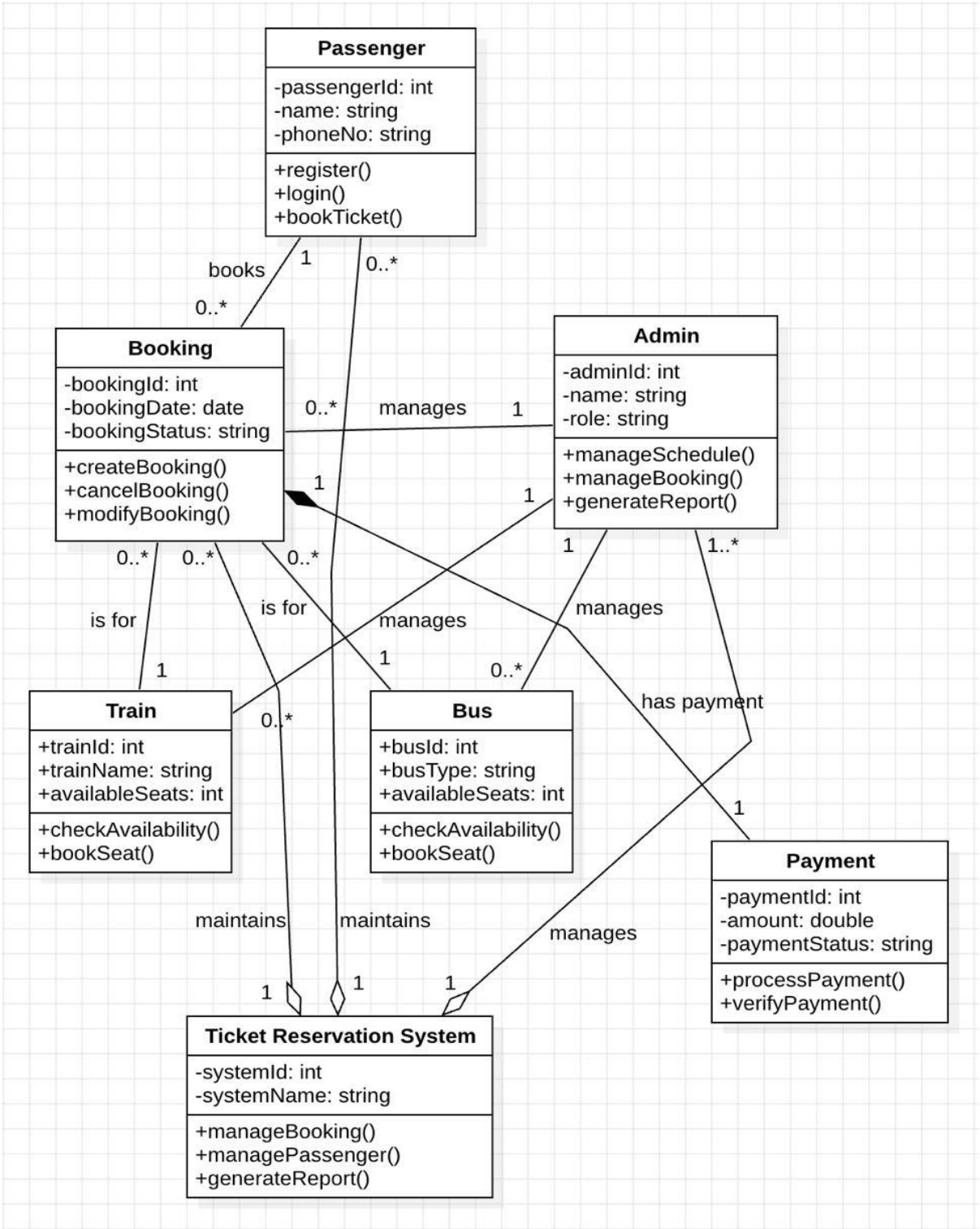
OOAD LAB ASSESSMENT-2
Ticket Booking and Reservation System

By
25MIS0355 Suhas E
25MIS0349 Nagasanjeevi Balaji N
25MIS0378 Lokesh Kumar G

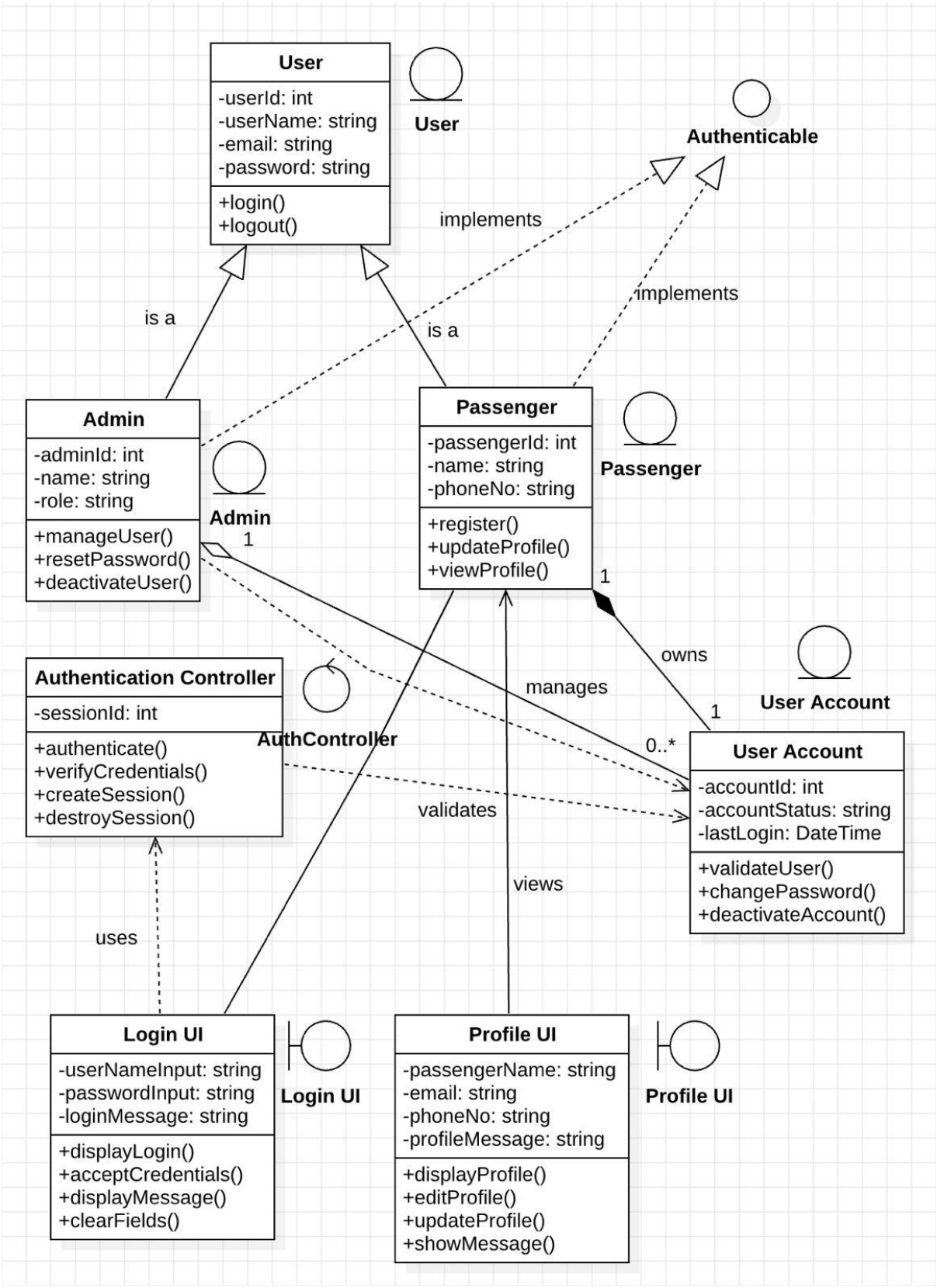
Overall Class Diagram:



Relations:

- **Composition**: Booking has a composition relationship with Payment, indicating that a payment cannot exist without a booking.
- **Aggregation**:
  - Ticket Reservation System has an aggregation relationship with Booking
  - Ticket Reservation System has an aggregation relationship with Passenger
  - Ticket Reservation System has an aggregation relationship with Admin
- **Association**:
  - Passenger has an association relationship with Booking, indicating that a passenger can make multiple bookings.
  - Admin has an association relationship with Booking, indicating that an admin manages multiple bookings.
  - Admin has an association relationship with Train, indicating that an admin manages multiple trains.
  - Admin has an association relationship with Bus, indicating that an admin manages multiple buses.
  - Booking has an association relationship with Train, indicating that multiple bookings can be made for a single train.
  - Booking has an association relationship with Bus, indicating that multiple bookings can be made for a single bus.

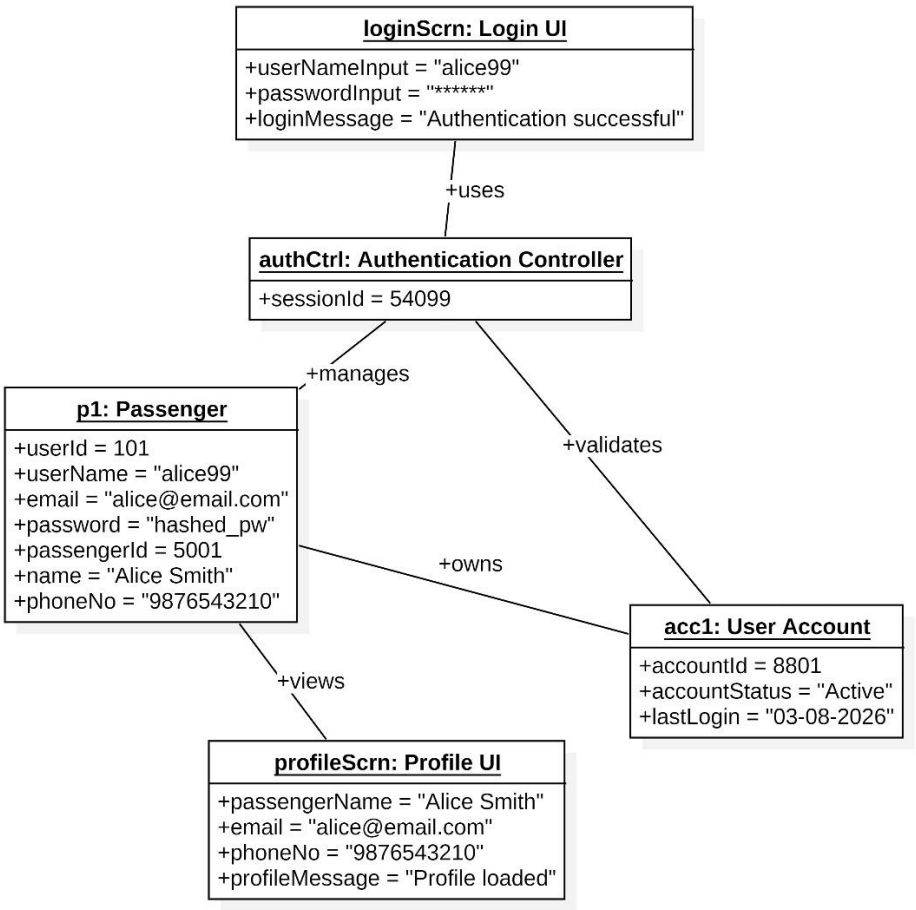
Module 1 – Login & User Management:



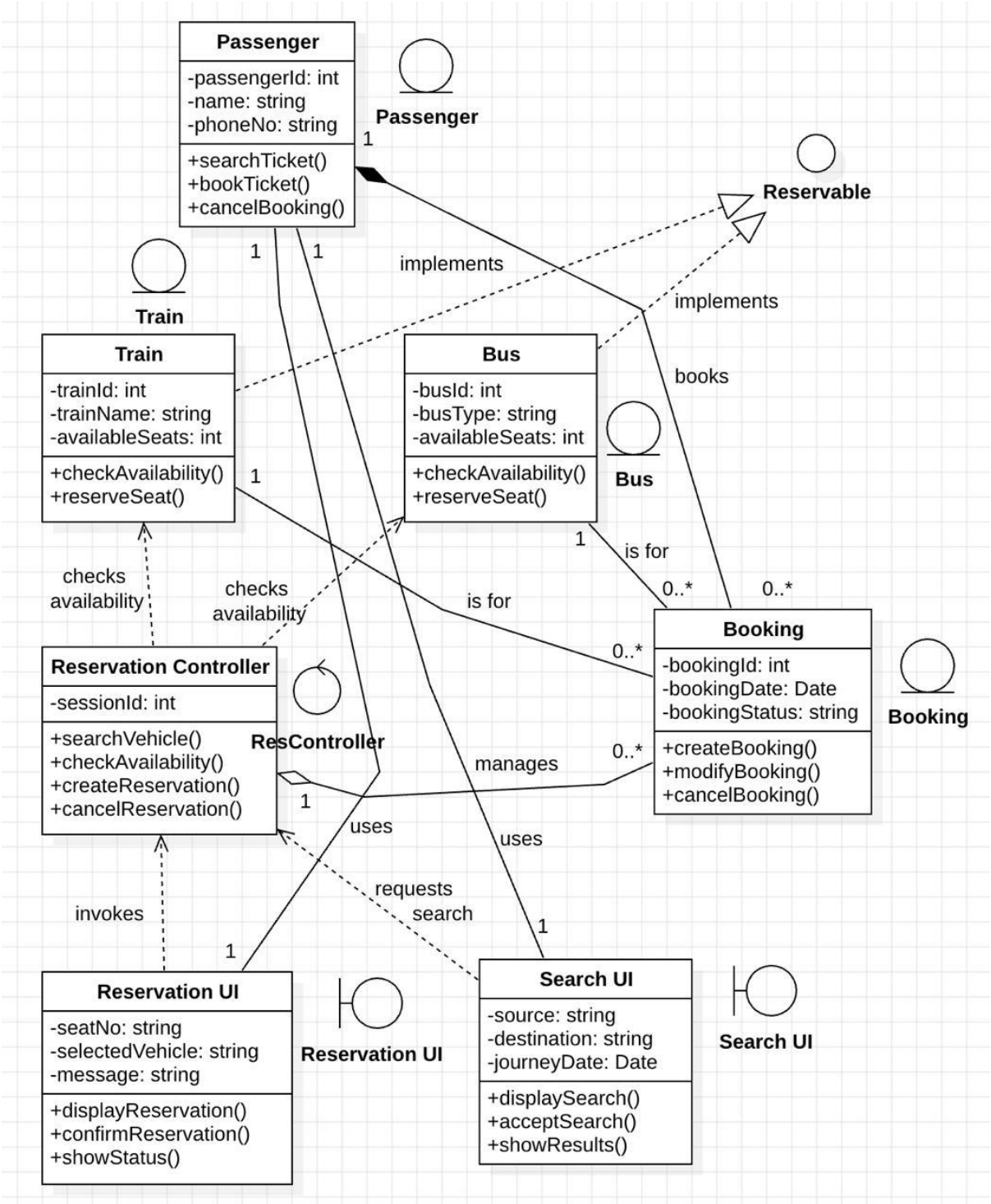
Relations:

- **Inheritance**: Admin and Passenger inherit from User.
- **Realization/Implementation**: Admin and Passenger implement the Authenticable interface.
- **Aggregation**: Admin has an aggregation relationship with User Account, indicating that an admin manages multiple user accounts.
- **Composition**: Passenger has a composition relationship with User Account, indicating that a user account cannot exist without a passenger.
- **Association**:
    - Login UI has an association relationship with Passenger, indicating that the passenger interacts with the login interface.
    - Profile UI has an association relationship with Passenger, indicating that the profile interface views the passenger data.
- **Dependency**:
    - Login UI depends on the Authentication Controller to authenticate and process credentials.
    - Authentication Controller depends on Passenger to manage user sessions.
    - Authentication Controller depends on User Account to validate credentials.

Object Diagram:



Module 2 – Ticket Search & Reservation: 25MIS0378

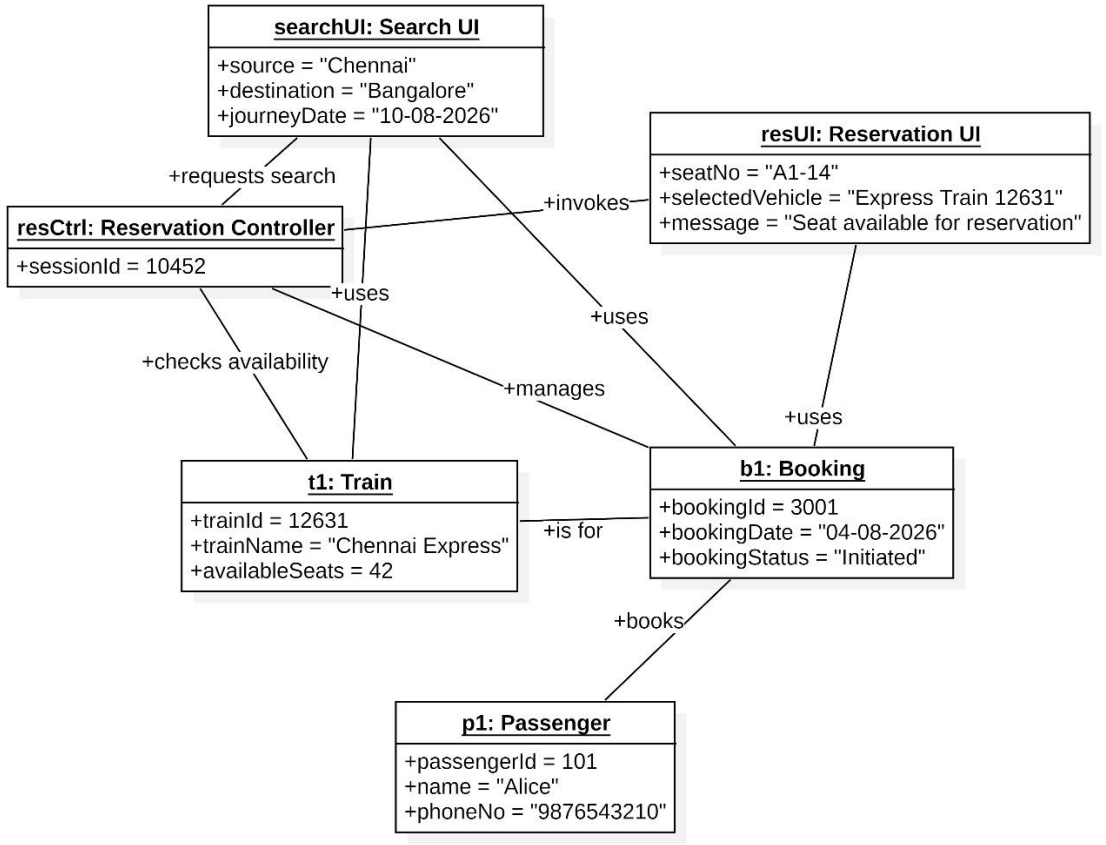


Relations:

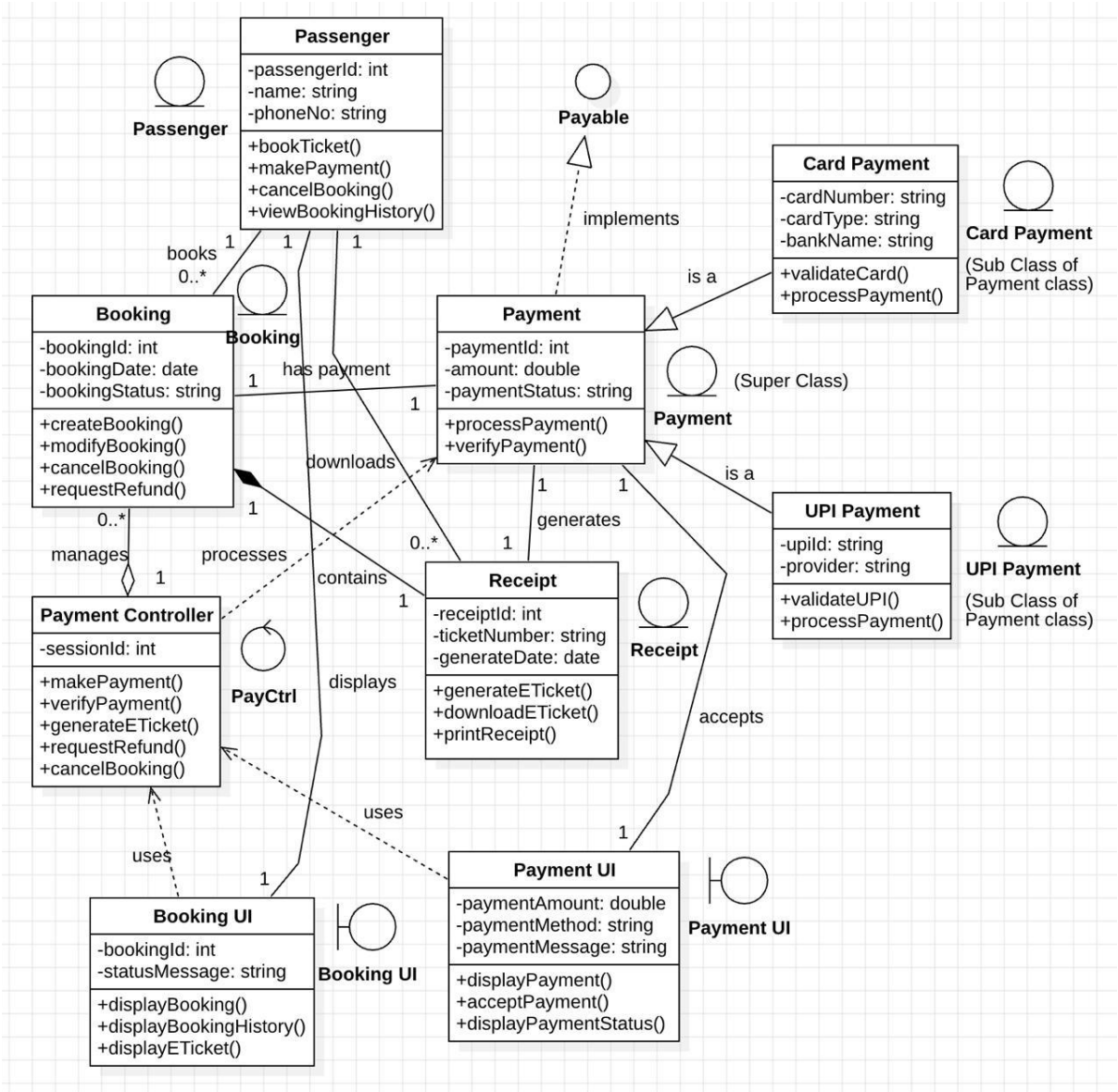
- **Realization/Implementation**: Train and Bus implement the Reservable interface.
- **Composition**: Passenger has a composition relationship with Booking, indicating that a booking cannot exist without a passenger.

- **Aggregation**: Reservation Controller has an aggregation relationship with Booking, indicating that the controller manages multiple bookings.
- **Association**:
    - Passenger has an association relationship with Search UI, indicating that the passenger uses the search interface.
    - Passenger has an association relationship with Reservation UI, indicating that the passenger uses the reservation interface.
    - Booking has an association relationship with Train, indicating that a booking is made for a specific train.
    - Booking has an association relationship with Bus, indicating that a booking is made for a specific bus.
- **Dependency**:
    - Search UI depends on the Reservation Controller to request a vehicle search.
    - Reservation UI depends on the Reservation Controller to invoke reservation logic.
    - Reservation Controller depends on Train to check vehicle availability.
    - Reservation Controller depends on Bus to check vehicle availability.

Object Diagram: 25MIS0378



Module 3 – Payment & Booking Management: 25MIS0349



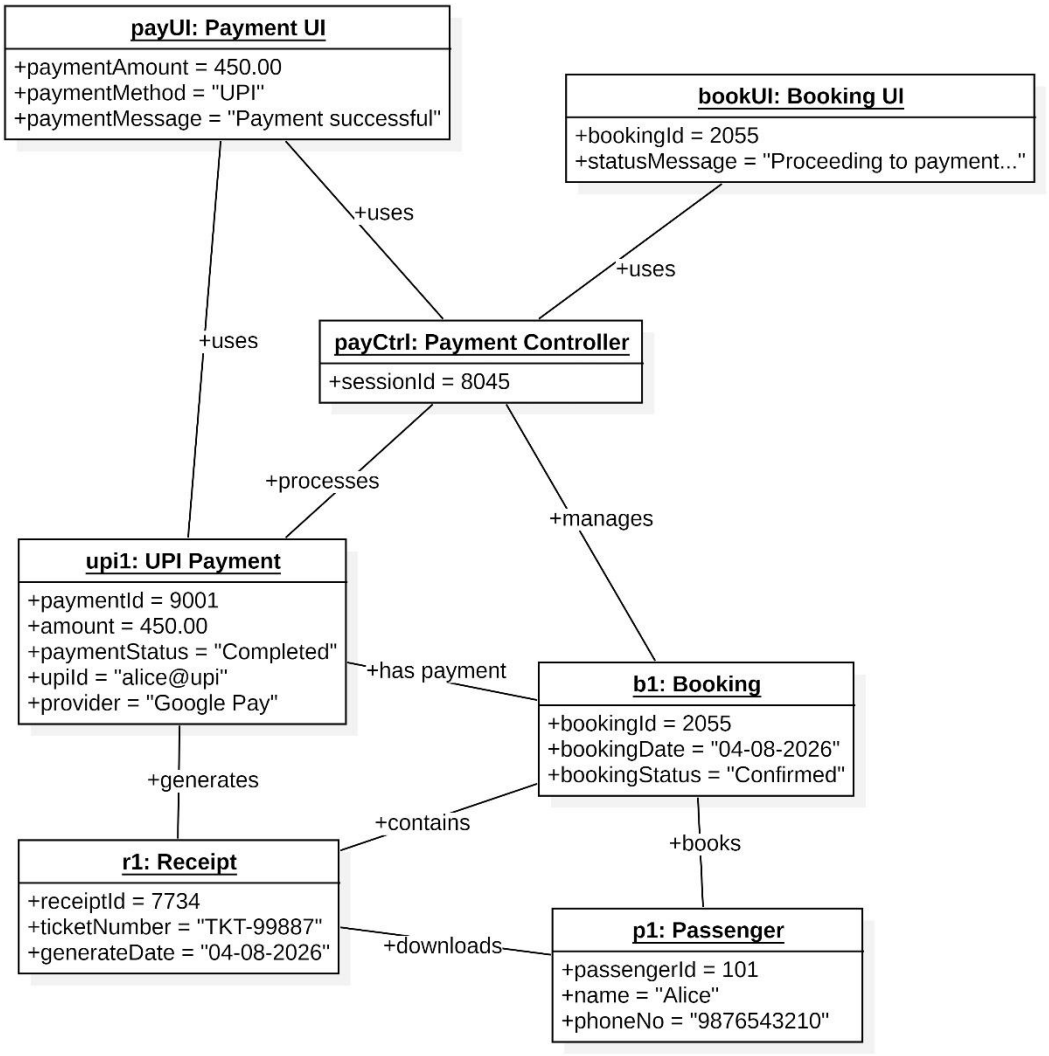
Relations:

- **Inheritance**: Card Payment and UPI Payment inherit from Payment.
- **Realization/Implementation**: Payment implements the Payable interface.
- **Aggregation**: Payment Controller has an aggregation relationship with Booking, indicating that the controller manages multiple bookings.
- **Composition**: Booking has a composition relationship with Receipt, indicating that a receipt is contained by and cannot exist without a booking.
- **Association**:
  - Passenger has an association relationship with Booking, indicating that a passenger can make multiple bookings.
  - Passenger has an association relationship with Receipt, indicating that a passenger downloads their generated receipt.

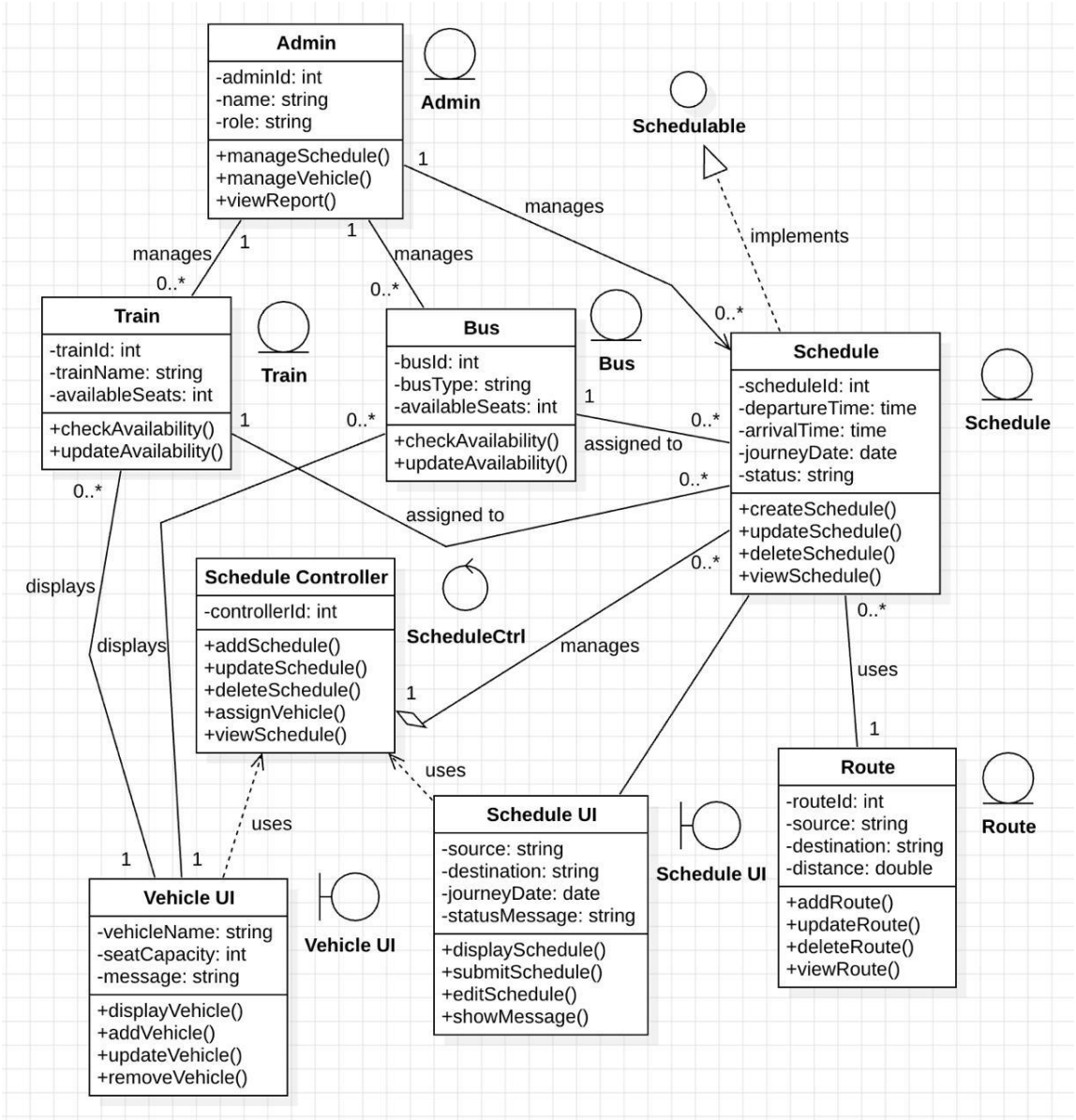
- Passenger has an association relationship with Booking UI, indicating that a passenger is displayed the UI.
- Booking has an association relationship with Payment, indicating that a booking has a corresponding payment.
- Payment has an association relationship with Receipt, indicating that a payment generates a receipt.
- Payment UI has an association relationship with Payment, indicating that the interface accepts the payment details.

- **Dependency**:
  - Booking UI depends on the Payment Controller (a "uses" relationship) to route the user to payment processing.
  - Payment UI depends on the Payment Controller (a "uses" relationship) to submit the payment details.
  - Payment Controller depends on Payment to process the actual transaction.

Object diagram: 25MIS0349



Module 4 – Admin & Schedule Management: 25MIS0355



Relations:

- **Realization/Implementation**: Schedule implements the Schedulable interface.
- **Aggregation**: Schedule Controller has an aggregation relationship with Schedule, indicating that the controller manages multiple schedules.
- **Association**:
  - Admin has an association relationship with Train, indicating that an admin manages multiple trains.
  - Admin has an association relationship with Bus, indicating that an admin manages multiple buses.
  - Admin has an association relationship with Schedule, indicating that an admin manages multiple system schedules.
  - Vehicle UI has an association relationship with Train, indicating that the interface displays train details.

- Vehicle UI has an association relationship with Bus, indicating that the interface displays bus details.
- Schedule has an association relationship with Train, indicating that a schedule is assigned to a specific train.
- Schedule has an association relationship with Bus, indicating that a schedule is assigned to a specific bus.
- Schedule has an association relationship with Route, indicating that a schedule uses a specific route.
- Schedule has an association relationship with Schedule UI.

- **Dependency**:
  - Vehicle UI depends on the Schedule Controller (a "uses" relationship) to process vehicle updates.
  - Schedule UI depends on the Schedule Controller (a "uses" relationship) to submit schedule changes.

Object Diagram: 25MIS0355

